UbiLearn: Supporting English-as-a-Foreign-Language Learners in Reflecting on Conversations Using a Smartwatch

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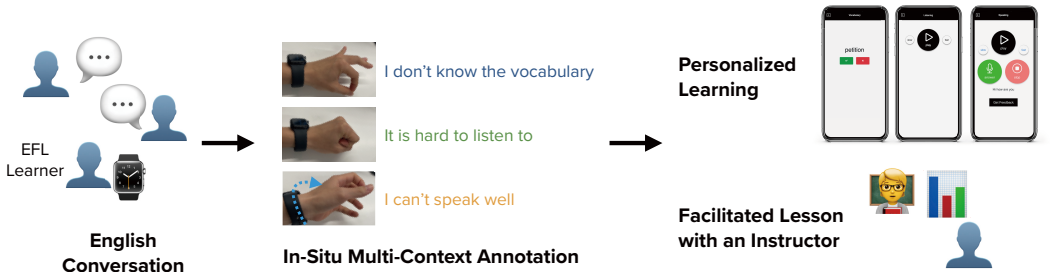


Fig. 1. UbiLearn helps English-as-a-Foreign-Language (EFL) learners tackle the difficulties in nurturing their English skills by enabling them to review conversational moments (*e.g.*, group conversation with fluent speakers). Powered by subtle hand gesture recognition on a smartwatch, it allows in-situ annotation for multiple contexts during the conversation without breaking its flow. The annotation is then processed with speech and natural language processing to create personalized educational material for learners to review. Moreover, it enables EFL instructors to understand each learner's in-situ proficiency, offering effective tutoring sessions.

What new opportunities can the current ubiquitous computing and AI technologies provide to support English-as-a-Foreign-Language (EFL) learners? To answer the question, we began with a formative study with EFL learners, uncovering multiple challenges during conversations with others and their desire to review such scenes later. We implemented a smartwatch prototype, *UbiLearn*, which features hand gesture recognition for in-situ multi-context annotation to save moments when learners face difficulty. The annotation is used to generate personalized educational material powered by speech and natural language processing. Through a series of studies, we demonstrated the feasibility and preferred usability of UbiLearn, leading to learners' enhanced learning satisfaction. Moreover, the annotation data promoted the role of instructors by enabling the tracking of learners' in-situ proficiency outside their tutoring sessions. We conclude by highlighting emerging opportunities for learners enabled by mobile and AI technologies, along with key considerations.

CCS Concepts: • **Human-centered computing** → **Interactive systems and tools**; **Ubiquitous and mobile computing systems and tools**; *Empirical studies in interaction design*.

Additional Key Words and Phrases: personalized learning, in-situ annotation, foreign language learning, smartwatch

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1 Introduction

Learning a second language, especially *English as a Foreign Language* (EFL) [27], is crucial in today's globalized era to foster foundational communication skills [79]. A significant number of studies in the HCI domain have explored this topic. Among them, ubiquitous computing is a powerful technology that provides pervasive learning opportunities to overcome the forgetting curve [18, 65, 82]. For instance, *VocabEncounter* [5] seamlessly integrates personalized foreign vocabulary into text users see in daily life through mobile phones or AR glasses. Prior studies emphasize the effectiveness of pervasive learning experiences to maintain long-term retention [18, 30, 39, 55], particularly in synchronizing learning with the user's experience and context.

However, a noticeable gap is that most learning interactions predominantly occur during limited, individual user experiences, such as reading texts [5] or watching videos [92]. Therefore, these approaches seldom connect with actual conversational scenes. The gap becomes crucial for EFL learners in English-speaking countries since they face challenges in conversations, struggling with understanding, formulating responses, or overcoming nervousness [58]. While solo rehearsal or traditional 1-on-1 tutoring can serve as a training opportunity, these challenges specifically reside during spontaneous, multi-speaker interactions [11]. To advance the frontier of HCI research in supporting EFL learners, this work aims to design a system that helps them leverage their spontaneous conversations for learning. We expect that such systems can be realized given the recent surge in wearable devices equipped with microphones and the advancements in on-device speech processing capabilities [62], heralding a new era of ubiquitous learning [12].

In this study, we primarily focused on international students in English-speaking countries as EFL learners, a demographic significantly affected by communication lapses, academic challenges, and mental health issues stemming from their difficulties in English conversations [85]. First, we conducted semi-structured interviews ($N = 10$) as a formative study to comprehend their challenges during English conversations, potential learning opportunities, and ideal interaction scenarios. Our findings highlighted a demand from EFL learners to revisit specific conversational instances for learning. Also, we revealed multiple contexts behind such instances, along with a preference among learners for non-disruptive, subtle interactions that do not interrupt the flow of conversation.

Based on the findings, we designed *UbiLearn*, a system enabling EFL learners to leverage their conversations for reflection anchored by in-situ multi-context annotation using a smartwatch (See Figure 1). Specifically, we catered to their need for unobtrusive annotations during conversations to save moments for further review without breaking the conversation flow. These annotations corresponded to three distinct contexts that were guided by the formative study: unfamiliarity with certain words (*vocabulary mode*), inability to distinguish what the other party says (*listening mode*), and struggles in articulating thoughts or pronouncing phrases (*speaking mode*). UbiLearn empowers users to annotate these moments during a conversation through subtle hand gesture recognition instantly with minimized effort using just a smartwatch. Then, after the conversation, users can use a tailored Web app that automatically yields personalized learning materials based on these in-situ annotations.

We first evaluated UbiLearn's technical feasibility, which confirmed that our system could recognize hand gestures made during conversations with a reasonable accuracy of 84.4%. Additionally, in the *vocabulary mode*, our system could accurately identify the word the user didn't understand from the recorded audio by using speech and natural language processing. Subsequently, we conducted

a user study involving EFL students ($N = 12$) conversing with fluent speakers. A comparison between scenarios using and not using UbiLearn showed that gesture interaction did not impose additional cognitive load during a conversation, while using the system resulted in greater learning satisfaction. Finally, to analyze the broader impact of UbiLearn and gain feedback, we conducted an in-the-wild study involving a tutoring session with five pairs of EFL learners and instructors. As a result, the system's positive influence beyond individual learning has been uncovered: enhancing the instructor-learner relationship by providing them with insights into learners' proficiency outside their tutoring sessions. We conclude the paper with a summary of findings and important considerations of ethics, privacy, and social acceptance of the emerging technology.

Our work contributes to the following:

(1) Design considerations for supporting EFL learners in reflecting on their conversations in English, which were informed by a formative study.
(2) UbiLearn, a novel smartwatch-based system built on top of subtle hand gesture recognition, speech processing, and natural language processing, which yields personalized learning materials on a tailored Web app.
(3) A user study that demonstrated the feasibility of UbiLearn and its preferred experience that leads to greater learning satisfaction compared to a conventional reviewing approach.
(4) Design implications of the in-situ multi-context annotation interaction in supporting EFL learners, evidenced through an in-the-wild study involving both learners and instructors.

2 Related Work

2.1 English as a Foreign Language (EFL)

Engaging in English as a Foreign Language (EFL) learning underpins communication among individuals in our progressively globalized era [79]. In particular, EFL learners with minimal English exposure during their formative years encounter various challenges [31, 85]. For instance, language barriers impact academic performance and permeate every aspect of daily life [8], especially for international students [58]. Consequently, many English learning materials and services have been proposed, such as smartphone applications and video materials that allow users to engage casually. Simultaneously, expanding learning opportunities where users can engage naturally within their daily lives without being confined to existing computing platforms is an essential research theme [12].

2.2 Systems to Practice Conversation in a Foreign Language

Computer-assisted language learning (CALL) has been extensively studied by researchers [6]. To facilitate conversation skills, previous studies have explored using computational agents as conversation partners with which learners can practice. Chatbots and speech processing techniques are widely used, such as [59, 60, 72, 77]. Wang *et al.* [83] proposed an educational robot as a tangible learning companion. Dizon [19] suggested the opportunity for EFL learners to use Amazon Alexa as a useful tool to enhance language learning.

While benefits are demonstrated, these CALL systems require learners to dedicate time to using them, which can hinder their continued use if they lose motivation. Moreover, these systems often use imaginary conversational situations based on the learner's interests and chosen topics. Furthermore, while they often assume one-on-one conversations for practice, the challenges of EFL learners are also associated with multi-party conversations [11]. Thus, a gap exists between the supported scenarios and the spontaneous, diverse conversations.

We take a contrasting approach by enabling EFL learners to learn from English conversations embedded in their own real-world, situated experiences. This setting introduces unique challenges

distinct from conventional learning systems, where learners typically engage at their own pace in controlled environments. In real conversations, learners cannot pause interactions and often struggle to review and reflect on them to effectively practice their English skills. To address these challenges, we investigate plausible interaction designs through iterative steps involving both EFL learners and instructors.

2.3 Systems to Support Pervasive Language Learning

In the context of CALL, offering learners with ubiquitous opportunities has been a favorable idea. HCI research has utilized such pervasive learning [30, 39], especially for vocabulary learning [5, 9, 16–18, 37, 53, 87, 92]. For example, Dingler *et al.* [17] placed pervasive physical displays in users' homes and work environments for vocabulary learning. *VocaBura* [37] is a smartphone app that enables EFL learners to study new vocabulary while walking. *Vivo* [92] is a video-augmented vocabulary dictionary users can use to study words or phrases while watching videos. The benefit of such pervasive learning is that users can learn new information within their life context.

Despite the benefits of CALL systems that offer pervasive learning opportunities, the situations supported by them have largely been limited to passive experiences, such as browsing or watching videos. In contrast, the most critical and practical context for language learning, *i.e.*, real-world conversation, remains relatively under-explored. We believe that such situated interactions provide rich opportunities for learning, allowing EFL learners to engage with and reflect on their actual use of language within the contexts they personally navigate.

2.4 Leveraging Life Logs and Its Concerns

Many studies have been conducted to utilize personal data such as life logs [33]. Recent advancements in mobile and wearable computing with audio processing capabilities, such as a smartwatch, AI Pin [40], and pendant [51], can introduce a new paradigm of utilizing everyday personal data for various applications. For instance, *Memoro* [94] is a wearable and audio-based personal memory assistant combining speech processing and Large Language Models (LLMs).

In these studies, researchers have explored ways to enable user input to provide context to systems [43, 70]. In response, there has been a growing interest in in-situ annotations within mobile computing, aiming to minimize the burden on users when reflecting on and annotating data retrospectively [67, 88, 91]. For instance, Tanaka *et al.* [80] designed a system that allows users to annotate important scenes in first-person videos using a single button. Researchers are investigating further ways to enhance the social acceptability by leveraging unobtrusive gestures [20, 45, 69]. For example, Exposito *et al.* [24] proposed *Unobtrusive Note Taking*, which augments the experience of note taking with foot-based gestures, and reported the benefit of its hands-freeness while engaged in conversation. Similarly, Cai *et al.* [10] used a small ring mouse to interact with other devices during conversation, reporting that the subtle interaction sustained the social acceptance.

On the other hand, privacy concerns have provoked a long-lasting discussion, especially when they involve the data collection of third parties [38, 73]. Given that, life-log research has presented several methods to automatically protect the privacy of users [71], such as by obfuscating personal information [48, 49] or by detecting moments not to be recorded [81]. In this regard, it is important to consider that such new technologies and their involving concerns have evolved over time. For example, Kärkkäinen *et al.* [41] reported the positive attitude of users toward a smartphone-based activity logging system as long as they have control over the logged content. Hoyle *et al.* [38] conducted an in-situ experiment in which 36 participants wore a life-logging camera for a week and reported that bystanders showed less concern than the participants wearing the camera expected. Also, when the users asked bystanders for consent, it was reported that most of them received positive interest, and no one encountered conflict. To further mitigate the privacy issue, recent

products (*e.g.*, Limitless AI [51]) employ speech processing, only recording those who have already verbally agreed with the recording.

Given such a trend, we expect that our scenario of leveraging conversations as a learning opportunity for EFL learners would not be impractical. More specifically, we use the recorded data for educational purposes alone and assume that other people in the conversation have agreed to its use in advance. We explore how such design fits into learners' lives and augments their learning experiences.

3 Formative Study

We conducted semi-structured interviews with international students to understand EFL learners' difficulties during English conversation. Our research interest, informed by the above background, lies in whether common devices can derive optimal learning experiences for EFL learners with the support of recent speech/language processing technologies.

3.1 Procedure

We invited ten volunteers (5 male, 5 female; aged between 21 and 32) through word-of-mouth for the semi-structured interviews. They are international students at English-speaking universities with less than four years of living experience. They self-reported a high level of proficiency in English, which was approximately equivalent to CEFR C1 [57].

We first asked questions about their experience of having conversations in English, especially the challenges, as well as how they had tried to overcome those. After that, we explained our assumption of recording conversations with pervasive devices, like a smartwatch, and utilizing the recorded data for retrospective reviewing. Then, we inquired about their acceptance and any expected benefits and difficulties that might occur when we build such a system. We explained the supposition that the recording would be only used for their own English practice, and others in the conversation agreed to it. Each interview took roughly 15 minutes.

3.2 Findings

The participants' responses were analyzed using open coding [78]. Through iterative refinement processes, we obtained the following topics.

3.2.1 Demand to review conversation. All participants agreed that they encounter difficulties in their conversation in English, such as:

> I sometimes struggle with casual conversations, especially in a group chat with people speaking English fluently. (P1)

> When I cannot catch up with the conversation, I must laugh by following others without understanding the context. Those moments are embarrassing, and I feel sad. I want to understand the conversation fully. (P3)

These comments reconfirm EFL learners' challenges during conversations in English while indicating their motivation to improve their skills. When asked how they had tried to overcome the difficulty, most of their answers were studying outside the conversation, such as reading vocabulary books and watching English videos. However, they mentioned the clear limitations of such approaches.

> Even if I study hard, connecting the study outcome with actual conversation remains challenging. In such cases, I often lose motivation. (P9)

This comment corroborates the benefit of the existing practice systems discussed in Section 2.2. Two participants mentioned that they also tried to practice within the ongoing conversation but acknowledged that it was not the optimal way; for example,

> I could sometimes ask the other person to repeat the sentence on the spot. However, it stops the conversation flow, especially if it is a group conversation, and I cannot do it frequently. (P4)

> I try to remember the words I am not familiar with during conversations to look up the meaning later, but in many cases, I forget them as the conversation continues. (P6)

These findings suggest a strong demand for retrospective learning support grounded in real-world conversations. A key design implication is that the system should allow users to revisit specific moments of difficulty with minimal disruption during the conversation. This calls for lightweight logging mechanisms that do not have to stop the ongoing conversation.

3.2.2 Preference but hesitation to in-situ annotation. Then, we introduced the idea that common devices such as a smartwatch or a pendant device offer such reviewing experiences, and asked about their impression. Here, we also explained that users could mark scenes they want to review through real-time annotations, such as by tapping a button on the device. The participants agreed that the annotation function is beneficial to review the conversation efficiently, saying,

> Since the conversation happens often, anywhere, I only need to review moments when I have difficulty. Recording all conversations is redundant and infeasible to review. (P7)

At the same time, they showed concern about how to make annotations in real-time during the conversation. Participants mentioned explicit ways like tapping a button on the screen may not be acceptable.

> It could be useful since asking "Pardon?" every time is also hard for me. But, ideally, I prefer more subtle ways to annotate moments so that others would not notice that I do so. (P2)

While users saw value in marking difficult moments, their hesitation around overt interactions implies the need for subtle, socially acceptable input methods. A key design implication is that in-situ annotation should be possible through discreet, eyes-free interactions, such as wrist gestures or implicit behavior cues, rather than screen taps, which may draw unwanted attention.

3.2.3 Dilemma between in-situ annotation frequency and review workload. Moreover, a few participants mentioned an expected dilemma regarding using the annotation function.

> I can imagine that I would annotate many scenes, and thus, it would become harder to review them later, hindering my use of the system. (P10)

> Annotating scenes sounds helpful, but just listening to the scenes is not very exciting and an easy way to study. I'd appreciate a more personalized and smarter way of studying based on the annotation I provide. (P9)

These comments reveal a tension between the desire to annotate frequently and the burden of reviewing too many annotations later. This suggests the system should go beyond raw audio review and instead offer intelligent summarization or prioritization of annotated segments. A potential design feature is personal educational materials based on the interaction history.

3.2.4 Multiple contexts for the difficulty. The participants also mentioned various contexts for the difficulty, such as unknown vocabulary, not being able to distinguish fast speech, and the inability to articulate their thoughts. These comments suggest that the difficulty EFL learners encounter during conversation is multi-context and can be categorized, which is aligned with prior research

in learning science and cognitive psychology [26, 61, 75]. As pointed out by Zhang and Mi [90], the nuance of such difficulties depends on each learner's experience and proficiency. The diversity in perceived difficulty, ranging from vocabulary to articulation, underscores the need for personalized support. A clear design implication is that the system should incorporate contextual labeling to tailor learning materials to the user's specific challenges. This could guide context-aware feedback or adaptive review modules.

3.3 Summary

We reconfirmed the challenges EFL learners have within their conversations in English. Moreover, the findings derived design considerations central to supporting EFL learners in conversation. These include: (1) enabling subtle and unobtrusive in-situ annotation; (2) reducing review burden through intelligent summarization; and (3) supporting context-aware personalization of learning materials. These principles directly informed our system design described in the following sections.

4 Prototype: UbiLearn

Based on the findings from the formative study, we developed *UbiLearn*, an interaction prototype using a smartwatch to learn from conversations by enabling users to perform in-situ multi-context annotation. In this section, we describe the design details and the implementation.

4.1 Why Smartwatch?

Underpinning our system are the recent advancements in speech-processing technologies. Specifically, the current technologies can transcribe the content of speech more accurately than humans [35], and they have also acquired robustness to noise and reverberations [68]. Furthermore, researchers have worked on implementing state-of-the-art speech recognition models on wearable devices [29], which has been a challenging task [50]. This allows the development of mobile interactive systems [7, 62, 94], for example, to support people who are deaf or hard-of-hearing [62]. We aimed to leverage the technology to support EFL learners by designing a system that meets the considerations we discussed in Section 3.3.

Smartwatches are one of the most common wearable devices and can be worn easily in everyday life. It is less privacy-invasive and power-hungry than camera-based systems, such as AR glasses. Moreover, recent studies have demonstrated that a smartwatch can be an effective input platform by leveraging gesture sensing. For instance, assistive touch[1] has been deployed on Apple Watches for navigation, and an approach to customizing gestures has also been proposed [86]. This gesture recognition capability frees users from the limitations of the small screen, where selecting different buttons can be difficult [15]. This background motivated us to incorporate the unobtrusive input techniques discussed in Section 2.4 to enable EFL learners to subtly perform in-situ annotations and record audio.

4.2 System Overview

The system overview is presented in Figure 2. A learner wears a smartwatch that has our developed app. Once launched, our watch app runs in the background. The motion is continuously processed to detect hand gestures for annotating scenes. Different gestures can be recognized to enable them to annotate multiple contexts for the difficulty. Simultaneously, the watch has a short time buffer, saving audio data only when any gesture is recognized, instead of recording all the time. The saved audio data and recognized gestures are sent to our server and then processed together through speech recognition and a natural language processing pipeline to generate learning materials,

[1] https://support.apple.com/en-us/HT212760

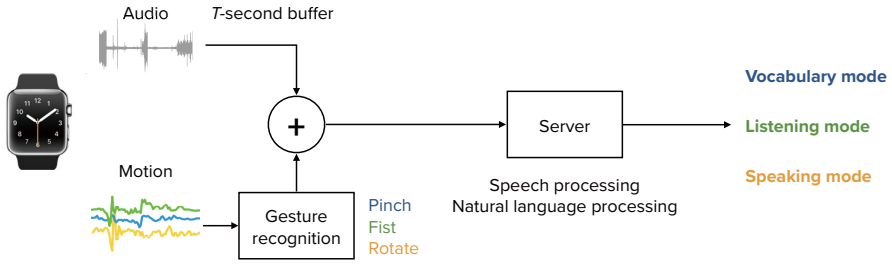


Fig. 2. Overview of the system architecture. The watch measures audio and motion with a T-second buffer for the audio. The audio buffer is saved with the recognized gesture information only when a hand gesture is detected. They are sent to the server together at the end of the conversation, where annotated data is processed to generate personalized learning materials.

which are personalized based on the learner's annotation. After the conversation, the learner can look up their smartphone or laptop to interact with the materials for their review.

4.3 Context-Aware Learning Mode

The formative study informed us of the importance of considering multiple contexts of difficulties EFL learners experience during a conversation. We involved an external English teacher in designing the user experience based on the interview results while referring to user interfaces in popular language learning apps. Through iteration, we finally developed three learning modes: *vocabulary mode*, *listening mode*, and *speaking mode*. Figure 3 presents the experience in each mode:

Vocabulary mode is prepared to help learners memorize unfamiliar words and phrases they encountered during conversations. As detailed in Appendix 4.5.3, it first determines the word or phrase most likely to be unknown around the moment when learners perform the corresponding gesture by using GPT-4. In addition to the word description from the Oxford English Dictionary, we also provide the contextual explanation of the word's meaning, supplied by the LLM. This pair of information can serve as a personalized vocabulary book in the web app.

Listening mode also detects the word or phrase that the learner would have had difficulty understanding, but presents the audio clip around the word, instead of presenting the word itself. Then, the user can repeatedly listen to the clip, by slowing down or speeding up if needed, along with the transcript.

Speaking mode is meant to practice formulating and voicing thoughts in similar conversational contexts. It plays back the audio clips from the moment learners made the corresponding gesture, and consequently, invites them to have in-the-moment *what-if* continuation. The real-time transcript of their speech is presented in the web app, and once they stop speaking, a high-level feedback for the conversation flow is provided using GPT-4.

Please refer to Appendix A for more detailed design and implementation strategies of each mode, including the prompts for GPT-4 that we designed with the EFL teacher to provide pedagogically appropriate materials. Note that we acknowledged that these categories are interrelated and not mutually exclusive; learners may encounter listening difficulty due to unknown vocabulary. Our intention is not to rigidly classify each challenge but to provide a scaffold that supports learners in applying different types of processing beyond simply listening to raw audio clips, as revealed in our formative study.



Fig. 3. Three learning modes implemented in the prototype of UbiLearn. (1) *Vocabulary mode*: learners can look up and try a quiz of annotated unfamiliar words and phrases identified from the saved audio clip. (2) *Listening mode*: learners can listen to the audio clips they had difficulty understanding with control of their speed and generated transcription. (3) *Speaking mode*: learners can practice articulating their thoughts in situations in which they had difficulty. The answer is transcribed, and feedback is generated by GPT-4. The detailed usage is described in Appendix A.

4.4 Gesture-Based In-Situ Multi-Context Annotation

As we discussed, the smartwatch can be an input platform to enable learners to perform in-situ annotations. In this regard, as there are three learning modes, the system must detect three distinguished gestures. To decide on the mapping, we conducted a pilot study with three volunteers, starting with a set of seven gestures by considering prior work [84]: *pinch*, *double pinch*, *tap*, *double tap*, *fist*, *rub fingers*, and *rotate*. In this process, we focused on user cognitive load to perform different gestures in real time during conversations based on different contexts. As a result, we found that

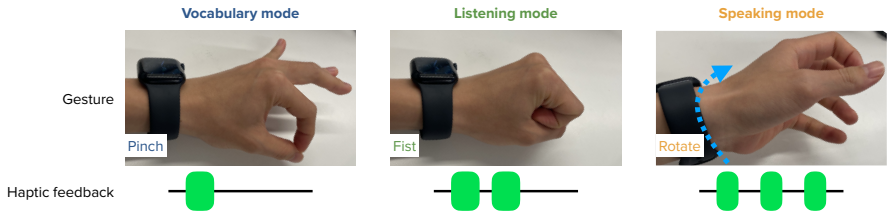


Fig. 4. The finalized mapping of learning mode, hand gesture, and haptic feedback.

the gestures should be distinguishable enough to prevent confusion both in terms of user error and sensing error. For example, *pinch* and *tap* are similar gestures, and pilot users mentioned they were confused. Also, we found that gestures containing double actions (*double pinch* and *double tap*) and lasting finger movements (*rub fingers*) impose more physical load than other gestures, potentially leading to their fatigue. Thus, we decided to use *pinch*, *fist*, and *rotate* gestures.

When we asked the pilot users to think of a mapping between the three learning modes and the three hand gestures, we noticed their tendency to use *pinch* for *vocabulary mode*, *fist* for *listening mode*, and *rotate* for *speaking mode*. It was mentioned that the *pinch* gesture can be associated with the idea of catching a word or phrase in an utterance. Also, fist can be deemed as a larger action than pinch, and such a relationship matches with the context of *vocabulary mode* and *listening mode*, indicating that not knowing vocabulary is a part of not distinguishing the utterance. Moreover, *rotate* is slightly different from other gestures in the sense that it rotates the wrist instead of fingers, which also matches the relationship of *speaking mode* to the other two modes as it is the only mode expected to occur at/after the user's own utterance. While preliminary, we thought such intuitiveness would help users perform gestures smoothly, as demonstrated in prior gesture-based interface [56]. Thus, we decided to adopt the above mapping for the subsequent study. Note that such a mapping would still be easily customizable [86] when the app is deployed to many end users.

Regarding gesture detection, it is necessary to prevent fluctuating outputs that occur due to the frame-by-frame prediction of the model and confusion with other non-related hand movements. Therefore, we introduced a heuristic in the on-device processing of motion data. Specifically, we assumed that gestures occur when hands are still, which we call *relax* state. Detecting *relax* is relatively easy by looking at the total energy of motion data, which is cost-effective as it does not involve machine learning (ML) inference. Then, the gesture recognition model is applied to a 1-second window only when windows before and after the window of interest are detected as *relax*. In this way, we can exclude non-relevant hand movements and obtain stable predictions while avoiding frequent ML inference. We expected that such an interaction would be natural for most conversations when users are seated or standing. However, it excludes situations where the user's hands move constantly, such as walking, for which we need more sophisticated processing.

Lastly, a soft haptic feedback mechanism is demanded since users perform the annotation gestures while being engaged in the primary conversation. Users could not confirm whether their annotation was correctly recorded during conversations without such feedback. For this, we utilized the smartwatch's haptics feature. We assigned different haptics to each of the three gestures and made the smartwatch return the corresponding haptics. The finalized mapping of learning mode, gesture, and haptic feedback is summarized in Figure 4.

4.5 Implementation Details

4.5.1 Subtle gesture recognition on smartwatch. We used the Apple Watch Series 7 and developed a custom app using Xcode. Since we could not access the APIs for the assistive touch at the time of the prototype development, we trained our own ML model and deployed it on the watch using the CoreML framework[2]. To this aim, we collected data from ten volunteers. We asked them to perform the designated gestures (*i.e.*, *pinch*, *fist*, *rotate*, and *other*) one by one while recording IMU data at 50 Hz using Core Motion framework[3]. Here, *other* means any motion not similar to the other three gestures. The participants performed each gesture within a 1-second window. As a result, we obtained 1,215 instances. Through feature exploration, we used raw values of acceleration (xyz) and gyro (xyz) sensor values within the 1-second window, which is a feature of $50 \times 6 = 300$ values. We trained a Random Forest classifier (with default scikit-learn [66] parameters). Once trained, the model was converted into a CoreML format and loaded onto the Apple Watch. For the haptic feedback, we used the CoreHaptics framework[4], and the chosen haptics are presented in Figure 4. The gesture recognition module ran on the watch with an average latency of 0.2 seconds, fast enough for real-time annotation. In terms of battery performance, our app can run for approximately 8 hours, demonstrating its feasibility for continuous daily use.

4.5.2 Speech processing. On the watch, a $T = 10$-second audio buffer was used. This value was determined in our pilot study for gesture mapping as a reaction time for users to perform annotation when they encounter difficulty. The audio data in the buffer was stored as a separate *wav* file whenever a gesture was recognized. This approach could mitigate the privacy concern since it does not need to record the entire conversation. The separate audio files were uploaded to our server and then processed by a speech recognition pipeline consisting of Whisper [68]. Also, we applied speaker diarization [64] to separate the speech content of different speakers, which was then utilized by the natural language processing pipeline.

4.5.3 Text processing to identify unknown vocabulary. In the *vocabulary mode*, the transcript of the $T = 10$-second audio was then processed by the natural language processing pipeline to detect the exact word or phrase the user did not understand. For this purpose, we acknowledge that there are several methods to computationally model word difficulty in a personalized manner [21, 22]. Meanwhile, during our prototyping, we found that GPT-4 [63] performs well and decided to use it. We iterated on the system prompt, and the finalized prompt is presented in Figure 7. Here, we provided few-shot examples to stabilize outputs from GPT-4. We also enabled users to share their English level (*e.g.*, TOEFL score) upon user registration. If shared, the information was provided to GPT-4 and used to estimate word difficulties based on their English levels.

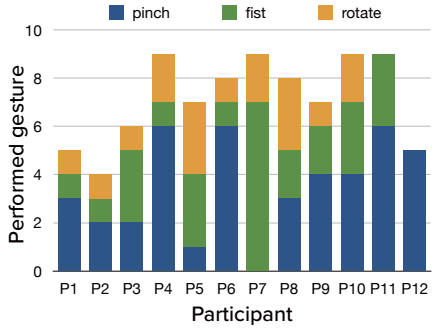
5 System Performance Evaluation

We first examined the performance of UbiLearn, that is, how accurately user annotations can be captured during conversation. In addition, for the *vocabulary mode*, we investigated the accuracy of the pipeline in detecting the phrases users had difficulty understanding.

5.1 Procedure

We recruited 12 EFL learners through word-of-mouth (6 male, 6 female; aged between 21 and 38) who self-reported that they sometimes faced difficulty in English conversation. Each learner had a roughly 7-minute conversation with fluent speakers while seated, and we asked them to annotate

[2] https://developer.apple.com/documentation/coreml
[3] https://developer.apple.com/documentation/coremotion
[4] https://developer.apple.com/documentation/corehaptics



Recognized gesture \ Performed gesture	Pinch	Fist	Rotate	Do not perform
Pinch	38	3	0	2
Fist	2	22	1	2
Rotate	0	0	16	0
Do not perform	2	2	0	N/A

Fig. 5. (left) Distribution of performed gestures across participants. (right) Results of the hand gesture recognition.

scenes whenever they faced difficulty. To examine the learner's behavior in different situations, we varied the group size between two and five, and each participant was randomly assigned to one size while balancing the distribution (*i.e.*, three participants for one group size). In the following, P1–P3 had a group size of two, P4–P6 had three, P7–P9 had four, and P10–12 had five. We did not specify the conversation topic. After an experimenter explained UbiLearn's concept and how to make annotations, they had a roughly 1-minute practice session to familiarize themselves with gesture recognition and haptic feedback. Before starting the session, we asked the learner to stop the conversation and let the experimenter know whenever they made a wrong gesture as opposed to their intention (*i.e.*, user error); for example, while they intended to perform *pinch*, they did *fist*. Every moment they performed a gesture, the experimenter observed it visually and took notes along with the timestamp to compare with the output with gesture recognition later. Also, if they performed the *pinch* gesture, the experimenter interrupted the conversation, asked which word or phrase they did not understand, and added their answer to the notes. No doubt, this approach created unnecessary pauses, but we prioritized accurately collecting ground truth user intention; we will evaluate the user experience in more ecologically valid settings in Study 1.

5.2 Annotations

The participants made 94 annotations in total. Out of them, eight gestures (~ 9%) were performed against their intention (user error). Six of them were the confusion of *pinch* and *fist* gestures. This result shows a reasonably low user error, suggesting the participants were unlikely to confuse the three gestures.

Figure 5 (left) shows the distribution of performed gestures across participants after removing these eight user errors. There is a variance in the context of EFL learners' difficulties during conversation. As expected in the design process of UbiLearn, *pinch* was the gesture performed most frequently overall, followed by *fist* and *rotate*. This result corroborates our design of assigning the gesture that requires the smallest movement (*i.e.*, *pinch*) to the *vocabulary mode* so that learners perform it instantly. Anecdotally, we can observe different behaviors among different group sizes. For example, when a learner was in a group of two (P1–P3), they made fewer annotations. This can be understood as that, in one-on-one conversation, a learner could understand the conversation more or could ask the other person to recite or rephrase naturally. On the other hand, the number of annotations increased when more fluent speakers joined the conversation, implying that the learner had difficulty catching up with their conversation. This trend is also supported by the

comments obtained in the formative study and implies that UbiLearn can be more helpful in group conversations.

5.3 Gesture Recognition Accuracy

Then, we compared human annotation and the output of the gesture recognition model. The confusion matrix is shown in Figure 5 (right). We noticed a few cases of false negative, where the performed gestures were not recognized (*i.e.*, the model did not run on the watch), or false positives, where non-related hand movement was recognized. In response, we added a row and column corresponding to the *do not perform* case. Overall, the accuracy of this three + one class classification was 84.4%. We conjecture that the low ratio of false recognition is because the participants followed our introduced heuristic well: gestures occur when the hand is in *relax* state. Moreover, the matrix indicates that *rotate* is rarely confused with other gestures by the model. This is matched with our design of assigning a slightly different learning mode (*i.e.*, *speaking mode*) to the gesture as discussed in Section 4.4.

5.4 Unknown Vocabulary Detection Accuracy

Lastly, we examined the accuracy of identifying vocabulary unknown to the learner. To this aim, we applied our *vocabulary mode* pipeline to the 42 moments corresponding to the *pinch* gesture performed by the participants and compared the outputs with what they had told the experimenter. As a result, 34 cases (81.0%) were detected correctly by our pipeline. Here, our pipeline sometimes outputs a phrase consisting of two or three words that include the target word. We considered such cases successful because the phrases can facilitate usage-based learning to increase the memorability of vocabulary [5, 53, 92]. The relatively high accuracy suggests that our system can identify vocabulary unknown to learners from the annotation.

Out of the eight failure cases, five were based on errors in speech recognition; in these instances, the voice was relatively small and masked by surrounding noise, or too fast or unclear. The other three were due to the limitations of our LLM-based approach. For example, if multiple words in the transcribed sentence are likely to be regarded as difficult for a given English proficiency, the prompted LLM struggles to identify the one matching the user's intent. These errors could affect user experience, so additional processing to increase the accuracy in both speech and natural language processing is desirable, as discussed later in Section 9.3.

6 Study 1: Individual Reflection after a Conversation

We conducted a user study to examine the usability of UbiLearn. We were interested in whether performing in-situ annotation could be a burden to users and how users feel about the overall learning experience. In Study 1, we followed the within-participant design with two conditions: *proposed* and *baseline*. In the *proposed* condition, the participants were provided with UbiLearn. After the conversation, they used UbiLearn's three learning modes to reflect on the conversation. In the *baseline* condition, they were not provided with the prototype, but the entire conversation was audio recorded upon participants' agreement. After the conversation, they were provided with the recorded audio file and its transcription generated by the same speech recognition pipeline with UbiLearn.

6.1 Procedure

We recruited 12 EFL learners (five male, seven female) aged between 24 and 38, with two participants choosing not to disclose their age. They self-reported a high level of proficiency in English, which was approximately equivalent to CEFR C1, and also mentioned that they sometimes face difficulty in English conversation. The overview of the study procedure is presented in Figure 6. Each participant

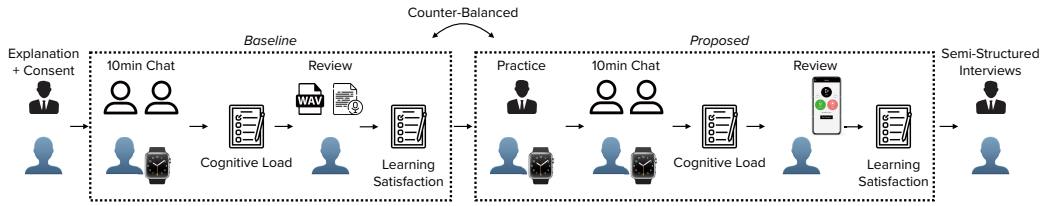


Fig. 6. Overview of the Study 1 procedure that uses a within-subject design comparing the *proposed* and *baseline* conditions.

had a face-to-face conversation with volunteer fluent speakers. We fixed the group size to three to see UbiLearn's efficacy in a group conversation, which our formative study and the performance evaluation indicated our system could be most beneficial for.

We asked the group (*i.e.*, the participant + two volunteers) to converse naturally without specifying the topic. Each conversation lasted approximately 10 minutes. In the *proposed* condition, we explained the usage of the system (*i.e.*, both gestures and interface) and asked them to use it naturally when they needed it. The participants had a 1-minute practice session to become familiar with gesture interactions. After each conversation, the participants completed a questionnaire measuring their cognitive load of having the group conversation under the assigned condition. Then, they reviewed the conversation by following the assigned condition. The participants finished the review at their own pace and then filled out a questionnaire corresponding to the learning satisfaction of the reviewing experience. The order of the two conditions was counter-balanced. Lastly, after they experienced both conditions, we conducted semi-structured interviews regarding the experience.

6.2 Metric

We measured the time taken to complete the self-review process in each condition. In addition, we measured participants' cognitive load using NASA-TLX [36] to examine the burden imposed by the proposed interaction of in-situ multi-context annotation during the conversation. The measurement employs six factors: mental demand, physical demand, temporal demand, effort, performance, and frustration. Also, we adopted the five-item questionnaire for measuring learning satisfaction that was validated in prior studies for evaluating satisfaction with language learning courses [34, 47]. We used 7-point Likert scale for the questionnaire, which consists of the following items: "This approach increased my interest in practicing English conversation," "I felt I achieved the objectives in this reviewing approach," "I liked the reviewing approach format," "I felt comfortable in this reviewing approach," and "I would recommend this reviewing approach to others."

6.3 Results

There were 12 data points for each factor of NASA-TLX [36]; we conducted the paired t-test after observing their normality. As a result, we did not see a significant difference for each factor: mental demand ($M = 4.0$, $SD = 1.7$ for the *baseline* and $M = 3.9$, $SD = 1.8$ for the *proposed*, $t(11) = 0.23$, $p = 0.82$, Cohen's $d = 0.046$), physical demand ($M = 2.4$, $SD = 1.3$ for the *baseline* and $M = 3.0$, $SD = 1.2$ for the *proposed*, $t(11) = -1.7$, $p = 0.11$, Cohen's $d = -0.45$), temporal demand ($M = 3.5$, $SD = 1.9$ for the *baseline* and $M = 3.1$, $SD = 1.7$ for the *proposed*, $t(11) = 0.77$, $p = 0.46$, Cohen's $d = 0.22$), effort ($M = 4.2$, $SD = 2.0$ for the *baseline* and $M = 4.1$, $SD = 1.6$ for the *proposed*, $t(11) = 0.12$, $p = 0.91$, Cohen's $d = 0.045$), performance ($M = 2.7$, $SD = 1.6$ for the *baseline* and $M = 2.4$, $SD = 1.3$ for the *proposed*, $t(11) = 0.61$, $p = 0.55$, Cohen's $d = 0.17$),

and frustration ($M = 2.5,\ SD = 1.4$ for the *baseline* and $M = 2.3,\ SD = 1.4$ for the *proposed*, $t(11) = 0.62,\ p = 0.55,$ Cohen's $d = 0.11$). This result suggested that the in-situ annotation using subtle hand gestures does not lead to a remarkable user burden.

We then analyzed learning satisfaction and the time taken to complete the self-review process in each condition. Similar to the analysis of NASA-TLX, there were 12 data points, and we conducted the paired t-test after checking normality. The result indicated that the *proposed* condition led to a significantly higher satisfaction ($M = 2.7,\ SD = 0.47$ for the *baseline* and $M = 3.8,\ SD = 0.66$ for the *proposed*, $t(11) = -3.7,\ p = 0.0034,$ Cohen's $d = -1.8$). Note that the five-item scale for learning satisfaction showed strong internal consistency for the full sample (Cronbach's $\alpha = 0.89$) and for the proposed condition ($\alpha = 0.90$), whereas this was marginal for the baseline condition ($\alpha = 0.66$). This implies that, while the baseline condition exhibited a more heterogeneous experience, UbiLearn resulted in a coherent and consistent positive evaluation.

In contrast, there was no significant difference in the time taken to finish reviewing the conversation between the conditions ($M = 237.4,\ SD = 154.4$ for the *baseline* and $M = 161.4,\ SD = 55.4$ for the *proposed*, $t(11) = 1.83,\ p = 0.09,$ Cohen's $d = 0.63$). In the *proposed* condition, we observed that all participants completed all quizzes in the *vocabulary mode* and *listening mode*. Additionally, four participants listened to the same audio at different speeds in the *listening mode*, and four repeated the speaking practice based on the feedback in the *speaking mode*. These results indicate that the learning material was well-designed, enhancing the subjective learning experience without significantly increasing the review time.

In addition, the number of gestures performed was 34, 23, and 11 for the *vocabulary*, *listening*, and *speaking modes*, respectively. The number of gestures performed per session was $M = 5.7,\ SD = 2.7$, which was slightly fewer than the number in Section 5, while the popularity of the gesture type remained consistent. We conjectured that this reflects a more natural usage of UbiLearn, as the participants in Section 5 were asked to perform gestures whenever they faced difficulty, likely leading to more frequent annotations. All participants interacted with each learning mode, in particular, by trying the quiz page. Still, this behavior could be due to the experimental setting (*i.e.*, their first-time use of the system), and a longer-time study is desirable to investigate how their usage shifts over time, as discussed later in Section 9.2.

6.4 User Comments

The results of the semi-structured interviews provided more qualitative insights into the experience UbiLearn enables. The participants' responses were analyzed using open coding [78].

6.4.1 Enhanced motivation to leverage the conversation. The participants favored UbiLearn's interaction as it affects their mindset during the group conversation by offering a way to annotate scenes.

> When I realize I don't understand what someone is saying, instead of feeling negative about it, I now see it as an opportunity to learn, allowing for a positive shift in perspective. (P6)

Since the learner actively drives the annotation, it enhances their awareness of practicing English during the conversation. EFL instructors also emphasized this point, which we will discuss later in Study 2.

6.4.2 Feelings about gesture-based annotation and its errors. The participants agreed that the gesture-based annotation is not a significant labor for them, corroborating the result of the cognitive load.

> The subtle gestures were easy to perform, especially when I listened to the others. I just paid a little attention to my left hand. (P1)

Some noticed sensing errors, such as when they did not feel haptic feedback (false negative) or extra feedback (false positive). Interestingly, they were not annoyed by false positives, likely thanks to the editing feature on the learning material.

> I sometimes felt feedback when I did not make gestures, probably because I made a similar motion unintentionally. When I looked at the summary page, I could easily spot them and mark them unimportant. In this sense, it is fine if extra detection happens. (P2)

Some participants expressed their feelings against false negative cases.

> At first, I felt a bit sad when the watch did not recognize my gestures, and I did not feel haptic feedback. Still, when I look back, I am not very disappointed since such moments were also not leveraged previously, and there would be plenty of opportunities to learn made possible by this tool if I were to use it throughout the day. (P6)

These comments suggest that, while it is important to continue improving the accuracy for a better experience, the proposed reviewing approach is robust against sensing errors because learners can easily correct false positive/wrong recognitions later based on their memory. Moreover, given that the enabled learning opportunities would be ubiquitous, learners would not regret even if they missed a few of them due to false negatives.

6.4.3 Feedback on each learning mode. The participants also gave feedback on each learning model.

> In the *vocabulary mode*, sometimes the identified vocabulary did not match my memory. (P9)

This was possibly due to the limitation of the current approach to identifying the unknown vocabulary, as we found in the performance evaluation, which requires additional work for improvement.

> I feel the accuracy in the *vocabulary mode* is satisfactory. It is exciting to see new phrases accumulated as a personalized dictionary. I would like to see what this will look like when I keep using it. For example, if I annotate the same phrase again, I want to get notified to strengthen my memory. (P5)

> I appreciate the *listening mode*; I can listen to the audio clips slowly and calmly. If I understand the meaning myself, I can gain confidence. However, sometimes, it remains challenging for me to grasp, for example, when the audio quality is too low, or it is beyond my ability. It would be great if the system would tell me if it's understandable or not. (P4)

> The speech recognition in the *speaking* mode allows me to check if my pronunciation is correctly recognized. As for the generated feedback, it is good if it relates to the grammatical and vocabulary points, such as letting me know alternative ways of saying similar stuff. It doesn't have to assess the content because I know the context better and do it myself. (P1)

These points inform us of the further room to refine the prototype while confirming the fundamental benefits of the designed interaction. Overall, Study 1 verified the design of offering personalized learning that is aware of the context behind the difficulty learners face during real-world conversations.

7 Study 2: Reflection with Instructors after In-the-Wild Use

Study 1 demonstrated the preferable experience of UbiLearn for individual reflection after a single conversation. We next explored the practical application of our system in real-world settings, particularly focusing on how it supports EFL instructors during their tutoring sessions. In Study 2,

we recruited five pairs of learners (L1 — L5) and instructors (I1 — I5) via word-of-mouth. Each learner had previously taken part in Study 1, with a minimum one-month gap between the two studies. None of the pairs had a previous tutoring relationship, and we matched them randomly.

Initially, each pair participated in a 30-minute tutoring session. Subsequently, we introduced the learners to our prototype and web application. Following this introduction, the learners had a three-day trial period. The learners were encouraged to use the system at their own pace. We instructed them to handle the system with the same caution as any audio recording app. After the trial, pairs engaged in another 30-minute tutoring session using the Web app. We did not prescribe a specific structure for the session, allowing each instructor to manage it independently. To conclude the study, we conducted semi-structured interviews with both learners and instructors separately to gather insights and feedback on their experiences.

7.1 Learners' In-the-Wild Usage

We aimed to comprehend how learners utilized the system and their perceptions of it in their lives. Learners indicated that they primarily used the system during conversations with their close friends. They found it challenging to use the recording function in other contexts due to the difficulty of introducing it to others. Interestingly, they also discovered the system's value in other scenarios to learn from spoken English, such as listening to the radio while driving or watching television shows.

> I think it was valuable, especially when I used it with native-speaker friends. I could catch new vocabulary, and listen to their conversation to understand what was talked about when they were speaking too fast for me. But what was actually helpful for me was when I used it myself engaged in listening to English, like TV shows. (L2)

> It was really convenient to use this when I was doing something else like cleaning a room or driving a car to attend English learning. The ability to review what I hear later is a helpful feature for learning. (L4)

They also mentioned that the custom app on the watch currently needs to be turned on manually before use (*i.e.*, Apple Watch does not allow a background recording for a long time), which heightened the hurdle for realizing the concept of ubiquitous learning. Based on their comments, we concluded that it is still challenging today to leverage all of our daily conversations for reviewing, from the perspective of social acceptance and technological limit. However, their comments also suggest that our system is certainly a step toward extending learning opportunities by enabling learners to review conversations with trusted people or to save spontaneous listening scenes, which are yet to be exploited by prior work discussed in Section 2.3.

7.2 Learners' Enhanced Motivation between Tutoring Sessions

When we asked about their experience with the tutoring session, the participants echoed our findings in Study 1, especially enhanced motivation for learning.

> Having this app, I tried to engage more in English communication. Gathering vocabulary and expressions was fun and fulfilling. I was looking forward to meeting the tutor to share my everyday issues, which I could not describe to them well otherwise. (L3)

> I usually meet my tutor once per week for a personal lesson but I actually don't study much outside the session. This system is great to track my everyday learning easily and can be a good pacemaker between tutoring sessions. (L1)

They also commented on the instructors' support in facilitating the use of UbiLearn.

> In our session, I could ask about what I could not distinguish in listening, and my instructor pointed out pronunciation patterns I should practice, which was nice. (L5)

These comments suggest that the saved data in UbiLearn facilitated the tutoring session and positively affected learners' learning attitudes between the sessions.

7.3 Evidence-Based, Contextualized Tutoring Session

All instructors appreciated the design of UbiLearn that offers insight into learners' proficiency with more contexts, saying,

> It was really helpful to have personalized material in our session because we could proceed with its contents first, like checking vocabulary or testing the speaking mode together, and then I could suggest the next steps and goals. (I2)

> From the recording in the listening mode, I could guess how the student was struggling with fast-paced conversations, which is really different from the English content we use in our ordinary sessions. In our case, they mentioned specific recordings that they could not distinguish, and I offered step-by-step examples. I believe that was really helpful. It would be great if the system itself could identify the pronunciation pattern, but it may not be reliable for now. (I1)

Their comments highlighted the UbiLearn' capability to offer insightful information by monitoring learners' conversational behavior with their friends or teachers, which would not be directly accessible to instructors otherwise. Observing the points at which a student hesitates or misuses language can highlight the need for real-time corrective feedback, helping them learn in the moment, such as in the example of I1.

Moreover, they expanded the idea to leverage the data further and advance personalized learning.

> I can imagine that this tool will transform our tutoring sessions in the future, especially when we have multiple sessions over time, and we can always check previous recordings to see how much we have advanced since then. That type of tracking or quantization will be motivating to students. (I2)

> The data will be valuable for us to track how the students' vocabulary zone changes over time through their everyday interactions. Knowing that is important for us to offer situated, personalized support. (I5)

One of UbiLearn's benefits is the ubiquitous observation of learners' English skills in a sustainable manner. Data-driven personalized language learning is an area of active research [46, 76], and UbiLearn could become a novel data source.

7.4 Effects on Instructor-Learner Relationship

Finally, they mentioned the broader effect UbiLearn could enable.

> Understanding the student's emotional responses to language barriers, like frustration or anxiety can inform us of features that make learning more supportive and less stressful. (I4)

Addressing the actual difficulties that a learner experiences during language acquisition can significantly enhance the relationship between the instructor and the learner. This approach not only focuses on the educational aspect but also on building trust and mutual respect, which are essential for effective learning.

> By creating an environment where difficulties are openly addressed without judgment, instructors encourage a culture of openness. Learners feel safe to express their vulnerabilities, which is crucial for addressing deeper learning barriers. (I1)

These comments suggest a novel form for the environment surrounding EFL learners facilitated by a ubiquitous intelligent system. Specifically, the overall comments present new learning opportunities that go beyond learners' isolated situations from instructors while leveraging their real-world conversations with the help of passive sensing and active annotation.

8 Discussion

Our series of studies validated the effectiveness of UbiLearn and its beneficial impact on learners. In Study 1, we discovered that UbiLearn increased learner satisfaction by enabling the review of individual conversations without adding cognitive burden. Study 2 revealed practical applications of UbiLearn, highlighting its various advantages for both learners and instructors. In this section, we synthesize our findings regarding EFL learners. Additionally, we discuss critical considerations concerning ethics, privacy, and social acceptance that are essential for the successful implementation of such technologies.

8.1 Insights about EFL Learner in English Conversation

This work sheds light on the multifaceted challenges, needs, and behaviors of EFL learners during real-world English conversations. Through both our formative study and performance evaluation, we observed that learners often struggle to follow group conversations, an area that remains underexplored in HCI, whereas one-on-one interaction support has been traditionally emphasized in the discipline (see Section 2.2). Notably, these challenges are not merely linguistic but also social in nature, often stemming from a hesitation to interrupt or disrupt the flow of conversation. This insight informed our design of a subtle, wearable-based annotation mechanism that preserves conversational naturalness while enabling learners to capture moments of difficulty.

Study 1 provided empirical evidence that enabling learners' in-the-moment engagement during conversation can enhance their overall participation and promote more active involvement. This aligns with findings from Galvan-Romero [28], who observed that intelligent personal assistants can increase learners' willingness to communicate, particularly when integrated into authentic, real-time interactions.

Our in-the-wild deployment further revealed that this positive effect was strengthened when instructors were involved in learners' review sessions in a *blended* format, as demonstrated by Arakawa and Yakura [4]. Echoing their findings, our results suggest that the instructor's presence boosts learner motivation and promotes deeper reflection between conversational episodes. This further supports prior work by Khan *et al.* [42], which highlights the importance of instructor scaffolding in the adoption of mobile learning tools. These findings also align with foundational theories of language learner autonomy and self-regulated learning [52, 93], which emphasize the value of learner agency, reflection, and structured support. Our system complements these frameworks by enabling reflection-on-action [74] through situated annotations, learner-controlled tools that encourage retrospective analysis of real conversational challenges. While blended learning formats have traditionally focused on integrating digital platforms into classroom instruction [44], our work extends this paradigm by embedding learners' spontaneous conversational experiences into the learning cycle, powered by wearable technologies.

Taken together, as discussed in Section 2.3, prior CALL research has successfully leveraged ubiquitous technologies to provide pervasive language learning opportunities, particularly for vocabulary acquisition through texts and videos (*e.g.*, [5, 92]). Building on this foundation, our findings demonstrate how recent advances in mobile computing and speech/language processing can turn everyday conversations into rich learning moments, offering new opportunities for reflective practice, and ultimately, fostering deeper and more sustained learner engagement.

8.2 Ethics and Privacy

As we discussed in Section 2.4, ethics and privacy should be carefully considered; recording other people without their knowledge is illegal in some countries and states. Similar to the work by Zulfikar *et al.* [94], our target scenario is where users gain consent from other speakers in advance, such as conversations with friends or instructors who are willing to support their learning. We keep carefully incorporating ways to balance technology's utility and user privacy before extending the scenario. For example, to prevent the recording of the speech from unexpected people (*e.g.*, nearby passengers), real-time sound separation [54] should be applied on the device. Products like Limitless AI [51] combine beamforming to extract the voice of those who have already verbally agreed with the recording. Similar techniques can be implemented in UbiLearn. Moreover, the raw audio files need to be discarded from the watch. Enabling the *vocabulary mode* only is a straightforward approach. Also, the *listening mode* could be modified to identify specific pronunciation from the recording that is difficult for the learner and present the information only, as suggested by I1 in Study 2. It is important to adjust the pipeline appropriately based on usage scenarios.

8.3 Social Acceptance

It is essential to consider the perceptions of other consenting participants in the conversation. Research indicates that the use of mobile devices during conversations is often viewed as impolite and distracting [1]. Cai *et al.* [10] used a small ring mouse to interact with other Internet-of-Things devices during conversation, reporting the subtle interaction sustained the social acceptance. Our design strategy is aligned with theirs by enabling more discreet interactions than traditional note-taking. Feedback from the speakers who interacted with the participants using UbiLearn in Study 1 implied minimal distraction or annoyance. However, this might not generalize to all situations. At this moment, we argue that the current design of UbiLearn should be limited to controlled settings where all participants are informed about its functionality and supportive of its use, as confirmed to be the most plausible cases in Study 2. At the same time, allowing non-native speakers to fully leverage their learning opportunities is important to mitigate the inequality caused by language barriers [13]. Together with ethics and privacy, social acceptance should be carefully considered before we extend the application scenarios in the future.

9 Limitations and Future Work

The current implementation of UbiLearn and study results are not without limitations.

9.1 Investigating the Effectiveness with Diverse Populations

First, the participant population was small, and a larger sample is necessary to rigorously evaluate the learning outcomes. For example, although there was at least a one-month interval, the fact that participants in Study 2 had previously taken part in Study 1 may have introduced prior bias. Moreover, our participants were mostly limited to EFL learners who already have a certain level of English proficiency while experiencing difficulty. Specifically, the level of the participants in our studies is approximately equivalent to C1 in CEFR [57], an advanced level of proficiency. People with different skill levels are supposed to use UbiLearn differently, and thus, additional studies with a more diverse population are demanded to examine the designed interaction.

Moreover, we tested the system with American English only, the population whose speech can be transcribed reasonably well with the state-of-the-art speech model [32]. However, the heterogeneity in speech, such as accent and clarity, influence the performance of speech recognition. As discussed in Section 5.4, the fast and unclear speech of fluent speakers was transcribed incorrectly, affecting the subsequent vocabulary detection. While research has been intensively conducted to address the

heterogeneity for speech recognition [23, 25], UbiLearn's experience needs to be evaluated with more diverse speakers.

This point is also applicable in regards to the diversity of the conversational contexts. As we use LLMs to produce part of the learning materials, we cannot fully guarantee that every content or feedback aligns with best-practice instructional sequencing or avoids subtle misconceptions, for example, when the conversation drifts into culturally nuanced topics or other domains that stretch beyond textbook English. Building a safeguard that remains effective across such open-ended contexts is therefore an open problem that contributes to more scalable deployment of LLM personalization in language learning. In addition, exploring its applicability to other languages, particularly those with different grammatical structures and phonetic systems, is equally important. Our future work includes such studies to achieve an inclusive technology.

9.2 Examining Learning Outcome

While Study 1 supported the favored user experience and its positive effect on their motivation, quantifying its learning effect requires more non-trivial efforts. For instance, it would require considering the differences in conversation, such as topics and the amount of speech between the control and treatment groups. Longer-term studies are desired to examine how the usage of UbiLearn shifts and how it affects learners' behavior and the learning outcome (*e.g.*, administering speaking assessment), as done by Chou *et al.* [14]. Additionally, as noted in Section 4.3, the three difficulty categories for EFL learners overlap. Our present work has not yet quantified the extent of these interdependencies, making a systematic evaluation of how UbiLearn boosts learners' skills and confidence across categories a critical avenue for future research. Still, our work demonstrating the preferable experience of the novel interaction from both the learner's and instructor's perspectives produces important design implications, paving the way for such a large-scale study in the future.

9.3 Improving Technical Components in the System

The current prototype has room for improvement. First, UbiLearn depends on the quality of the speech processing (*i.e.*, speech recognition) as suggested in Section 5.4. Therefore, conversations in noisy environments are unsuitable for using the system. Similarly, transcription quality is anticipated to decrease if the smartwatch is completely covered by clothing. To mitigate the issue, we can use additional processing, such as speech enhancement [89] and sound separation as front-end to speech recognition [3]. Secondly, the current gesture recognition assumes that gestures occur when hands are still. Thus, the system will struggle to recognize gestures while users are moving, such as walking. Therefore, additional processing to improve gesture recognition is demanded. At the same time, we plan to improve the learning app by enabling users to switch the category easily, not just to address the misclassification but also to facilitate their staged learning. As we discussed in Section 4.3, the categorized difficulties are interrelated; for instance, users might want to first train their vocabulary, proceed to the *listening mode*, and finally practice their speaking fluency to the same sentence. Providing such a global control is an important design, as recommended by Amershi *et al.* [2]. Lastly, the system uses GPT-4 to identify unknown vocabularies, which means users may be unable to use the system for low-resource languages for which LLMs are not developed. In such cases, we would rely on other approaches to identify a difficult word in a sentence. Generalizing our framework to other languages will be important to achieve inclusive technology, as discussed in Section 9.1.

10 Conclusion

We designed a novel system that enables EFL learners to learn effectively from their real-world conversations in English, which is a learning opportunity unexploited by prior HCI research. The

formative study sheds light on their challenges and needs in English conversations, informing key design considerations for the interaction, such as subtlety not interrupting the ongoing conversation. We implemented a prototype, UbiLearn, using a common smartwatch powered by subtle hand gesture recognition, speech processing, and natural language processing. Our user study demonstrated the preferable user experience of the proposed system and technical feasibility, highlighting the positive effect of in-situ annotation on learner satisfaction and motivation. Moreover, its in-the-wild use, followed by tutoring sessions, revealed further benefits of reflecting on conversational scenes they experienced with help from instructors, leading to evidence-based tutoring sessions and enhanced instructor-learner relationships. The results and derived design implications will be informative to researchers who use mobile technology to enhance the learning opportunities of EFL learners.

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A Details of the System

A.1 Vocabulary mode

Vocabulary mode offers two experiences: summary and quiz. In this mode, the words or phrases learners had difficulty understanding are identified from the recorded audio clips (the technical detail is described in Section 4.5.3). The prompt used for GPT-4 [63] to identify unknown vocabulary words from a given transcription is presented in Figure 7.

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You are an excellent English coach. Your student is showing a pretense of not understanding the content of a conversation. Your job is to guess what exactly is not understood. You should just specify a word or phrase and do not add any additional explanations or comments. Note that the TOEFL score of the student is [score].

Which word or phrase of the following conversation does the student not understand?
—
Speaker 0: and then it adjusts the whole thing on the fly.
Speaker 1: Dude,
Speaker 1: it's pig-ignorant.
Speaker 0: And it basically memorizes this lap,
Speaker 0: and then it beats any human on this lap.
—
Answer: pig-ignorant

Which word or phrase of the following conversation does the student not understand?
—
[transcript]
—
Answer:
```

Fig. 7. The prompt UbiLearn used to find a word or phrase that a user was unfamiliar with.

The summary page displays four pieces of information: identified phrases, their meaning, the number of times each phrase has been detected, and the last time it was detected, providing a vocabulary book-like interface for effective word study. Here, in addition to the explanation from the Oxford English Dictionary, the contextual explanation of their meaning is dynamically provided by GPT-4 using the prompt presented in Figure 8. A bookmark button is next to each phrase, enabling users to keep track of their entries. On the other hand, the quiz page tests phrases randomly chosen from the list. For each phrase, the user clicks on one of two check and cross-mark buttons to indicate whether they understand the phrase. Upon clicking the buttons, the phrase's correct meaning and the transcript of the T-second audio are presented, which would facilitate the user's usage-based learning as mentioned in Section 2.3.

A.2 Listening mode

Listening mode offers summary and quiz experiences. The audio clips learners had difficulty distinguishing can be reviewed in this mode. The summary page lists the audio clips, the time the clip was saved, and the bookmark button, similar to the *vocabulary mode*. In addition, the learner can add notes to each clip, such as making notes on words whose pronunciation was unclear. The quiz page plays an audio clip randomly chosen from the saved clips. The playback speed can be

You are an excellent English coach. Your student seems not to understand a specific word or phrase in a conversation. Your job is to tell the student the meaning of the word or phrase in the context of the conversation in a very concise manner. You should just tell a concise meaning of the word or phrase and do not add any additional explanations or comments nor recite the word or phrase.

What does “pig-ignorant” mean in the context of the following conversation?
—
Speaker 0: and then it adjusts the whole thing on the fly.
Speaker 1: Dude,
Speaker 1: it’s pig-ignorant.
Speaker 0: And it basically memorizes this lap,
Speaker 0: and then it beats any human on this lap.
—
Answer: extremely stupid or uneducated

What does “`[identified_phrase]`” mean in the context of the following conversation?
—
`[transcript]`
—
Answer:

Fig. 8. The prompt UbiLearn used to provide the contextual explanation of the phrase’s meaning for the *vocabulary* mode.

adjusted by two buttons (0.8x and 1.2x). Once the clip is finished, the user can note whether they comprehend it. Then, an automatically-generated audio transcript is presented, and the user can listen to it again if they want.

A.3 Speaking mode

Speaking mode also offers summary and quiz. The summary page is the same as the *listening mode*. On the quiz page, the user can hear an audio clip randomly chosen, and then they are given time to speak to the app to make a response. Here, they can see a real-time transcription of their speech. When the user finishes recording the response, they can press the “get feedback” button. Here, their audio is transcribed with Web Speech API[5], and the transcribed text is evaluated by GPT-4 [63] with a custom prompt that we obtained under the supervision of the English teacher. We expected that, while the feedback gives insights into their logical flow and vocabulary of the response, the transcription allows them to review their pronunciation, which was informed by the comment obtained in the formative study (See Section 3.2.4). The prompt we used for GPT-4 is presented in Figure 9.

[5]https://www.google.com/intl/ja/chrome/demos/speech.html

You are an excellent English coach. Please provide feedback on Speaker 1's response to Speaker 0's remark with regard to their use of English. The feedback format should start with a brief compliment, then a line break, followed by no more than 40 words highlighting areas for enhancement or providing suggestions for improvement. If Speaker 1 has not provided a response, please provide a suitable example.
—
Speaker 0: `[transcript]`
Speaker 1: `[new_response]`

Fig. 9. The prompt UbiLearn uses to give a user feedback about their response in the *speaking* mode.